

Visible envelope: version, epoch, header nonce, optional ratchet public material



sid, ibh, e, v
as AAD



Encrypted metadata: message index, envelope type, correlation id, payload nonce

mdk_e^{send}
metadata key



sid, ibh, e, n, v
as AAD



Encrypted payload: padded application plaintext

$mk_{e,n}$
message key

